Pandas program

1. **Create a Series**
    - Create a Pandas Series containing 5 student marks.
    - Display the Series and its index.

```
pands1.py - D:/107 Lab task/python/Pandas/pands1.py (3.14.7)
File  Edit  Format  Run  Options  Window  Help
import pandas as pd

marks_series = pd.Series([95, 62, 98, 97, 77],
                         index=['vivek', 'Mrudula', 'savitri', 'Divya', 'pratibha'])

print("--- Student Marks Series ---")
print(marks_series)

print("\n--- Series Index ---")
print(marks_series.index)
```

Output:

```
IDLE Shell 3.14.7
File  Edit  Shell  Debug  Options  Window  Help
Python 3.14.7 (tags/v3.14.7:823f032, Aug  5 2026, 10:51:32) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
================ RESTART: D:/107 Lab task/python/Pandas/pands1.py ================
--- Student Marks Series ---
vivek       95
Mrudula     62
savitri     98
Divya       97
pratibha    77
dtype: int64

--- Series Index ---
Index(['vivek', 'Mrudula', 'savitri', 'Divya', 'pratibha'], dtype='str')
>>>
```

2. **Create a DataFrame**
    - Create a DataFrame with columns: Name, Age, Marks.

- Add records for 5 students.
- Display the DataFrame.

pands2.py - D:/107 Lab task/python/Pandas/pands2.py (3.14.7)

File Edit Format Run Options Window Help

```
import pandas as pd

data = {
    'Name': ['Amruta', 'Samiksha', 'Rani', 'Disha', 'Kavita'],
    'Age':[22,21,23,22,21],
    'Marks': [83, 96, 88, 95, 78]
}
df = pd.DataFrame(data)
print("--- Student DataFrame ---")
print(df)
```

Output:

```
=============== RESTART: D:/107 Lab task/python/Pandas/pands2.py
--- Student DataFrame ---
      Name  Age  Marks
0    Amruta   22     83
1  Samiksha   21     96
2      Rani   23     88
3     Disha   22     95
4    Kavita   21     78
```

3. **Access Data**
   - Create a student DataFrame.

- Display only the Name column.
- Display the first 3 rows.
- Display the last 2 rows.

```
pands3.py - D:/107 Lab task/python/Pandas/pands3.py (3.14.7)
File  Edit  Format  Run  Options  Window  Help
import pandas as pd

data = {
    'Name': ['Devansh', 'Darshan', 'Saloni', 'Unati', 'Raviraj'],
    'Age':[22,21,23,23,21],
    'Marks': [85, 92, 78, 90, 88]
}
df = pd.DataFrame(data)

print("--- Name Column Only ---")
print(df['Name'])
print("\n" + "="*30 + "\n")

print("--- First 3 Rows ---")
print(df.head(3))
print("\n" + "="*30 + "\n")

print("--- Last 2 Rows ---")
print(df.tail(2))
```

Output:

```
========== RESTART: D:/107 Lab task/python/Pandas/pands3.py
--- Name Column Only ---
0    Devansh
1    Darshan
2     Saloni
3      Unati
4    Raviraj
Name: Name, dtype: str

==============================

--- First 3 Rows ---
      Name  Age  Marks
0  Devansh   22     85
1  Darshan   21     92
2   Saloni   23     78

==============================

--- Last 2 Rows ---
      Name  Age  Marks
3    Unati   23     90
4  Raviraj   21     88
```

4. **Add a New Column**

- Create a DataFrame containing Name, Maths, and Science.

- Add a new column Total containing the total mark

pandas4.py - D:/107 Lab task/python/Pandas/pandas4.py (3.14.7)

File Edit Format Run Options Window Help

```
import pandas as pd
data = {
    'Name': ['Arya', 'Snehal', 'Rasika', 'Divansha', 'Kanchan'],
    'Maths':[89,67,78,90,59],
    'Science': [90, 88, 82, 95, 84]
}
df = pd.DataFrame(data)

df['Total'] = df['Maths'] + df['Science']

print("--- DataFrame with Total Marks ---")
print(df)
```

Output:

IDLE Shell 3.14.7

File Edit Shell Debug Options Window Help

```
Python 3.14.7 (tags/v3.14.7:823f032, Aug  5 2026, 10:51:32) [MSC
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>

========== RESTART: D:/107 Lab task/python/Pandas/pandas4.py
--- DataFrame with Total Marks ---
      Name  Maths  Science  Total
0     Arya     89       90    179
1   Snehal     67       88    155
2   Rasika     78       82    160
3  Divansha    90       95    185
4  Kanchan     59       84    143
>>>
```

5. **Delete a Column**

- Create a DataFrame with Name, Age, City, and Marks.

- Delete the City column.

```
pandas5.py - D:/107 Lab task/python/Pandas/pandas5.py (3.14.7)
File  Edit  Format  Run  Options  Window  Help
import pandas as pd

data = {
    'Name': ['Arnab', 'Samridhi', 'Ramit', 'Divyam', 'Kritika'],
    'Age':[22,21,23,21,23],
    'City': ['Mumbai', 'Delhi', 'Bangalore', 'Pune', 'Chennai'],
    'Marks': [85, 92, 78, 90, 88]
}
df = pd.DataFrame(data)
df = df.drop('City', axis=1)

print("--- DataFrame After Deleting City Column ---")
print(df)
```

Output:

```
IDLE Shell 3.14.7
File  Edit  Shell  Debug  Options  Window  Help
Python 3.14.7 (tags/v3.14.7:823f032, Aug  5 2026, 10:51:32) [MSC v.
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: D:/107 Lab task/python/Pandas/pandas5.py =
--- DataFrame After Deleting City Column ---
      Name  Age  Marks
0    Arnab   22     85
1  Samridhi   21     92
2    Ramit   23     78
3   Divyam   21     90
4  Kritika   23     88
>>>
```

6. **Sort Data**

- Create a student DataFrame.

- Sort students according to their Marks in ascending order.
- Then sort in descending order.

pands6.py - D:/107 Lab task/python/Pandas/pands6.py (3.14.7)

File Edit Format Run Options Window Help

```
import pandas as pd

data = {
    'Name': ['Arnab', 'Samridhi', 'Ramit', 'Divyam', 'Kritika'],
    'Age':[21,23,22,21,23],
    'Marks': [85, 92, 78, 90, 88]
}
df = pd.DataFrame(data)
df_ascending = df.sort_values(by='Marks', ascending=True)

print("--- Sorted by Marks (Ascending) ---")
print(df_ascending)
print("\n" + "="*35 + "\n")

df_descending = df.sort_values(by='Marks', ascending=False)

print("--- Sorted by Marks (Descending) ---")
print(df_descending)
```

Output:

IDLE Shell 3.14.7

File Edit Shell Debug Options Window Help

```
Python 3.14.7 (tags/v3.14.7:823f032, Aug  5 2026, 10:51:32) [MSC
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
================ RESTART: D:/107 Lab task/python/Pandas/pands6.py
--- Sorted by Marks (Ascending) ---
      Name  Age  Marks
2    Ramit   22     78
0    Arnab   21     85
4   Kritika   23     88
3    Divyam   21     90
1  Samridhi   23     92

===================================

--- Sorted by Marks (Descending) ---
      Name  Age  Marks
1  Samridhi   23     92
3    Divyam   21     90
4   Kritika   23     88
0    Arnab   21     85
2    Ramit   22     78
>>>
```

7. **Filter Data**

- Create a DataFrame containing student names and marks.

- Display only students who scored **more than 70 marks**.

```
pandas7.py - D:/107 Lab task/python/Pandas/pandas7.py (3.14.7)
File  Edit  Format  Run  Options  Window  Help
import pandas as pd
data = {
    'Name': ['Aryan', 'Samridhi', 'Rakhi', 'Disha', 'Kirti'],
    'Marks': [85, 92, 68, 90, 55]
}
df = pd.DataFrame(data)
filtered_df = df[df['Marks'] > 70]
print("--- Students Scoring More Than 70 Marks ---")
print(filtered_df)
```

Output:

```
IDLE Shell 3.14.7
File  Edit  Shell  Debug  Options  Window  Help
Python 3.14.7 (tags/v3.14.7:823f032, Aug  5 2026, 10:51:32) [MSC v
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>

========== RESTART: D:/107 Lab task/python/Pandas/pandas7.py
--- Students Scoring More Than 70 Marks ---
      Name  Marks
0    Aryan     85
1  Samridhi     92
3    Disha     90
>>>
```

8. Create a DataFrame containing Name, Age, and Marks, with some missing values.
    - Find the missing values using Pandas.
    - Replace missing marks with 0.

pandas8.py - D:/107 Lab task/python/Pandas/pandas8.py (3.14.7)

File Edit Format Run Options Window Help

```
import pandas as pd
import numpy as np
data = {
    'Name': ['Arati', 'sara', 'Riya', 'diya', 'anu'],
    'Age': [20, np.nan, 21, 20, 19],
    'Marks': [85, 92, np.nan, 90, np.nan]
}
df = pd.DataFrame(data)

print("--- Original DataFrame with Missing Values ---")
print(df)
print("\n" + "="*45 + "\n")

print("--- Missing Values Matrix (isna) ---")
print(df.isna())
print("\n" + "="*45 + "\n")

df['Marks'] = df['Marks'].fillna(0)

print("--- DataFrame After Replacing Missing Marks with 0 ---")
print(df)
```

Output:

IDLE Shell 3.14.7

File Edit Shell Debug Options Window Help

```
Python 3.14.7 (tags/v3.14.7:823f032, Aug  5 2026, 10:51:32) [MSC v
Enter "help" below or click "Help" above for more information.
>>>
================ RESTART: D:/107 Lab task/python/Pandas/pandas8.py
--- Original DataFrame with Missing Values ---
    Name   Age  Marks
0  Arati  20.0   85.0
1   sara   NaN   92.0
2   Riya  21.0    NaN
3   diya  20.0   90.0
4    anu  19.0    NaN

=============================================

--- Missing Values Matrix (isna) ---
    Name    Age  Marks
0  False  False  False
1  False   True  False
2  False  False   True
3  False  False  False
4  False  False   True

=============================================

--- DataFrame After Replacing Missing Marks with 0 ---
    Name   Age  Marks
0  Arati  20.0   85.0
1   sara   NaN   92.0
2   Riya  21.0    0.0
3   diya  20.0   90.0
4    anu  19.0    0.0
>>>
```

9. **Student Result Analysis**

Create a CSV file named students.csv containing the following columns: RollNo, Name, Python, Java, DBMS
Store data for at least 10 students.

Using Pandas, read the CSV file and perform the following operations:

a) Display the complete dataset.
b) Display the first 5 records.
c) Calculate Total Marks for each student.
d) Calculate Percentage for each student.
e) Add a Result column:
Percentage >= 40 →
Pass Percentage < 40
→ Fail
f) Find the student who scored the highest percentage.
g) Find the average percentage of all students.
h) Display students who scored more than 75%.
i) Sort students according to percentage in descending order.
j) Save the updated data into student_result.csv.

student_analysis.py - E:/107 Lab task/python/Pandas/Student_Result/student_analysis.py (3.13.15)

File Edit Format Run Options Window Help

```
import pandas as pd

# Read the CSV file
df = pd.read_csv(r"E:\107 Lab task\python\Pandas\Student_Result\students.csv")

# a) Display the complete dataset
print("a) Complete Dataset:")
print(df)

# b) Display the first 5 records
print("\nb) First 5 Records:")
print(df.head(5))

# c) Calculate Total Marks for each student
df["Total"] = df[["Python", "Java", "DBMS"]].sum(axis=1)
print("\nc) Total Marks:")
print(df[["RollNo", "Name", "Total"]])

# d) Calculate Percentage for each student
df["Percentage"] = (df["Total"] / 300) * 100
print("\nd) Percentage:")
print(df[["RollNo", "Name", "Percentage"]])

# e) Add the Result column
df["Result"] = df["Percentage"].apply(
    lambda x: "Pass" if x >= 40 else "Fail"
)
print("\ne) Student Results:")
print(df[["RollNo", "Name", "Percentage", "Result"]])

# f) Find the student with the highest percentage
topper = df.loc[df["Percentage"].idxmax()]
print("\nf) Student with Highest Percentage:")
print(topper[["RollNo", "Name", "Percentage"]])

# g) Find the average percentage
average = df["Percentage"].mean()
print("\ng) Average Percentage:")
print(round(average, 2), "%")

# h) Display students who scored more than 75%
print("\nh) Students Scoring More Than 75%:")
print(df[df["Percentage"] > 75])

# i) Sort students by percentage in descending order
df = df.sort_values(by="Percentage", ascending=False)
print("\ni) Students Sorted by Percentage:")
print(df)

# j) Save the updated data into a new CSV file
df.to_csv("student_result.csv", index=False)
print("\nj) Updated data saved to student_result.csv")
```

Output:

IDLE Shell 3.13.15

File Edit Shell Debug Options Window Help

```
Python 3.13.15 (tags/v3.13.15:4061bc4, Aug  5 2026, 13:05:39) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
=== RESTART: E:/107 Lab task/python/Pandas/Student_Result/student_analysis.py ==
a) Complete Dataset:
   RollNo    Name  Python  Java  DBMS
0       1    Amit      80    75    85
1       2    Neha      90    88    92
2       3   Rahul      35    40    30
3       4   Priya      70    65    75
4       5      Om      50    55    60
5       6   Sneha      95    90    98
6       7   Rohan      25    30    35
7       8   Pooja      60    70    65
8       9   Akash      45    50    40
9      10  Anjali      85    80    90

b) First 5 Records:
   RollNo   Name  Python  Java  DBMS
0       1   Amit      80    75    85
1       2   Neha      90    88    92
2       3  Rahul      35    40    30
3       4  Priya      70    65    75
4       5     Om      50    55    60

c) Total Marks:
   RollNo    Name  Total
0       1    Amit    240
1       2    Neha    270
2       3   Rahul    105
3       4   Priya    210
4       5      Om    165
5       6   Sneha    283
6       7   Rohan     90
7       8   Pooja    195
8       9   Akash    135
9      10  Anjali    255

d) Percentage:
   RollNo    Name  Percentage
0       1    Amit   80.000000
1       2    Neha   90.000000
2       3   Rahul   35.000000
3       4   Priya   70.000000
4       5      Om   55.000000
5       6   Sneha   94.333333
6       7   Rohan   30.000000
7       8   Pooja   65.000000
8       9   Akash   45.000000
9      10  Anjali   85.000000

e) Student Results:
   RollNo    Name  Percentage Result
0       1    Amit   80.000000   Pass
1       2    Neha   90.000000   Pass
```

IDLE Shell 3.13.15

File Edit Shell Debug Options Window Help

```
0        1    Amit   80.000000
1        2    Neha   90.000000
2        3   Rahul   35.000000
3        4   Priya   70.000000
4        5      Om   55.000000
5        6   Sneha   94.333333
6        7   Rohan   30.000000
7        8   Pooja   65.000000
8        9   Akash   45.000000
9       10  Anjali   85.000000

e) Student Results:
   RollNo    Name  Percentage Result
0       1    Amit   80.000000   Pass
1       2    Neha   90.000000   Pass
2       3   Rahul   35.000000   Fail
3       4   Priya   70.000000   Pass
4       5      Om   55.000000   Pass
5       6   Sneha   94.333333   Pass
6       7   Rohan   30.000000   Fail
7       8   Pooja   65.000000   Pass
8       9   Akash   45.000000   Pass
9      10  Anjali   85.000000   Pass

f) Student with Highest Percentage:
RollNo                6
Name              Sneha
Percentage    94.333333
Name: 5, dtype: object

g) Average Percentage:
64.93 %

h) Students Scoring More Than 75%:
   RollNo    Name  Python  Java  DBMS  Total  Percentage Result
0       1    Amit      80    75    85    240   80.000000   Pass
1       2    Neha      90    88    92    270   90.000000   Pass
5       6   Sneha      95    90    98    283   94.333333   Pass
9      10  Anjali      85    80    90    255   85.000000   Pass

i) Students Sorted by Percentage:
   RollNo    Name  Python  Java  DBMS  Total  Percentage Result
5       6   Sneha      95    90    98    283   94.333333   Pass
1       2    Neha      90    88    92    270   90.000000   Pass
9      10  Anjali      85    80    90    255   85.000000   Pass
0       1    Amit      80    75    85    240   80.000000   Pass
3       4   Priya      70    65    75    210   70.000000   Pass
7       8   Pooja      60    70    65    195   65.000000   Pass
4       5      Om      50    55    60    165   55.000000   Pass
8       9   Akash      45    50    40    135   45.000000   Pass
2       3   Rahul      35    40    30    105   35.000000   Fail
6       7   Rohan      25    30    35     90   30.000000   Fail

j) Updated data saved to student_result.csv
>>>
```

10.Employee Salary Analysis

Create a CSV file named employees.csv containing: EmpID, Name, Department, Salary, Experience Store data for at least 10 employees.

Using Pandas, perform the following:

a) Read the CSV file.
b) Display all employees.
c) Display employees whose salary is greater than ₹40,000.
d) Find the highest salary.
e) Find the lowest salary.
f) Calculate the average salary.
g) Find the average salary of each department using groupby().
h) Display employees having more than 3 years of experience.
i) Sort employees by salary in descending order.
j) Save the result into employee_analysis.csv.

employee_salary.py - E:/107 Lab task/python/Pandas/employee_salary.py (3.13.15)

File Edit Format Run Options Window Help

```python
import pandas as pd

# Create employees.csv
data = {
    "EmpID": [101, 102, 103, 104, 105, 106, 107, 108, 109, 110],
    "Name": ["Amit", "Neha", "Rahul", "Priya", "Sneha",
             "Rohit", "Pooja", "Akash", "Kiran", "Sahil"],
    "Department": ["IT", "HR", "IT", "HR", "Sales",
                   "IT", "Sales", "HR", "Finance", "Finance"],
    "Salary": [55000, 35000, 45000, 60000, 42000,
               38000, 50000, 30000, 48000, 65000],
    "Experience": [4, 2, 5, 6, 3, 1, 7, 2, 4, 8]
}

df = pd.DataFrame(data)
df.to_csv("employees.csv", index=False)

# a) Read CSV file
df = pd.read_csv("employees.csv")

# b) Display all employees
print("a) All Employees:")
print(df)

# c) Salary greater than 40000
print("\nc) Salary Greater Than 40000:")
print(df[df["Salary"] > 40000])

# d) Highest salary
print("\nd) Highest Salary:")
print(df["Salary"].max())

# e) Lowest salary
print("\ne) Lowest Salary:")
print(df["Salary"].min())

# f) Average salary
print("\nf) Average Salary:")
print(df["Salary"].mean())

# g) Department-wise average salary
print("\ng) Average Salary by Department:")
print(df.groupby("Department")["Salary"].mean())

# h) Experience more than 3 years
print("\nh) Experience More Than 3 Years:")
print(df[df["Experience"] > 3])

# i) Sort salary in descending order
df = df.sort_values(by="Salary", ascending=False)
print("\ni) Sorted Employees:")
print(df)

# j) Save result
df.to_csv("employee_analysis.csv", index=False)
print("\nj) Saved to employee_analysis.csv")
```

Output:

IDLE Shell 3.13.15

File Edit Shell Debug Options Window Help

```
Python 3.13.15 (tags/v3.13.15:4061bc4, Aug  5 2026, 13:05:39) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: E:/107 Lab task/python/Pandas/employee_salary.py ==========
a) All Employees:
   EmpID   Name Department  Salary  Experience
0    101   Amit         IT   55000           4
1    102   Neha         HR   35000           2
2    103  Rahul         IT   45000           5
3    104  Priya         HR   60000           6
4    105  Sneha      Sales   42000           3
5    106  Rohit         IT   38000           1
6    107  Pooja      Sales   50000           7
7    108  Akash         HR   30000           2
8    109  Kiran    Finance   48000           4
9    110  Sahil    Finance   65000           8

c) Salary Greater Than 40000:
   EmpID   Name Department  Salary  Experience
0    101   Amit         IT   55000           4
2    103  Rahul         IT   45000           5
3    104  Priya         HR   60000           6
4    105  Sneha      Sales   42000           3
6    107  Pooja      Sales   50000           7
8    109  Kiran    Finance   48000           4
9    110  Sahil    Finance   65000           8

d) Highest Salary:
65000

e) Lowest Salary:
30000

f) Average Salary:
46800.0

g) Average Salary by Department:
Department
Finance    56500.000000
HR         41666.666667
IT         46000.000000
Sales      46000.000000
Name: Salary, dtype: float64

h) Experience More Than 3 Years:
   EmpID   Name Department  Salary  Experience
0    101   Amit         IT   55000           4
2    103  Rahul         IT   45000           5
3    104  Priya         HR   60000           6
6    107  Pooja      Sales   50000           7
8    109  Kiran    Finance   48000           4
9    110  Sahil    Finance   65000           8

i) Sorted Employees:
   EmpID   Name Department  Salary  Experience
9    110  Sahil    Finance   65000           8
3    104  Priya         HR   60000           6
0    101   Amit         IT   55000           4
```

```
2     103  Rahul          IT   45000           5
3     104  Priya          HR   60000           6
6     107  Pooja       Sales   50000           7
8     109  Kiran     Finance   48000           4
9     110  Sahil     Finance   65000           8

i) Sorted Employees:
   EmpID   Name Department  Salary  Experience
9     110  Sahil     Finance   65000           8
3     104  Priya          HR   60000           6
0     101   Amit          IT   55000           4
6     107  Pooja       Sales   50000           7
8     109  Kiran     Finance   48000           4
2     103  Rahul          IT   45000           5
4     105  Sneha       Sales   42000           3
5     106  Rohit          IT   38000           1
1     102   Neha          HR   35000           2
7     108  Akash          HR   30000           2

j) Saved to employee_analysis.csv
>
```

11.Sales Data Analysis

Create a CSV file named sales.csv
containing: Product, Category, Price,
Quantity
Store at least 10 product records.

Using Pandas, perform the following:

a) Read the CSV file.
b) Display all products.
c) Create a new column Total_Sales:
d) Total_Sales = Price × Quantity
e) Find the product with the highest total sales.
f) Find the total sales of all products.
g) Find the average product price.
h) Display products where quantity is greater than 10.
i) Find total sales category-wise using groupby().
j) Sort products according to Total_Sales in descending order.
k) Save the updated DataFrame into sales_analysis.csv.

sales_analysis.py - E:/107 Lab task/python/Pandas/sales_analysis.py (3.13.15)

File Edit Format Run Options Window Help

```
import pandas as pd

# Create sales data
data = {
    "Product": ["Laptop", "Mobile", "Mouse", "Keyboard", "Monitor",
                "Tablet", "Printer", "Headphone", "Speaker", "Charger"],
    "Category": ["Electronics", "Electronics", "Accessories",
                 "Accessories", "Electronics", "Electronics",
                 "Electronics", "Accessories", "Accessories", "Accessories"],
    "Price": [50000, 20000, 500, 1000, 12000,
              15000, 8000, 1500, 2500, 600],
    "Quantity": [5, 15, 25, 12, 8, 10, 6, 20, 14, 30]
}

df = pd.DataFrame(data)
df.to_csv("sales.csv", index=False)

# a) Read CSV file
df = pd.read_csv("sales.csv")

# b) Display all products
print("b) All Products:")
print(df)

# c and d) Calculate Total Sales
df["Total_Sales"] = df["Price"] * df["Quantity"]
print("\nc) and d) Total Sales:")
print(df[["Product", "Total_Sales"]])

# e) Product with highest total sales
print("\ne) Product with Highest Sales:")
print(df.loc[df["Total_Sales"].idxmax()])

# f) Total sales of all products
print("\nf) Total Sales:")
print(df["Total_Sales"].sum())

# g) Average product price
print("\ng) Average Product Price:")
print(df["Price"].mean())

# h) Products with quantity greater than 10
print("\nh) Quantity Greater Than 10:")
print(df[df["Quantity"] > 10])

# i) Category-wise total sales
print("\ni) Category-wise Total Sales:")
print(df.groupby("Category")["Total_Sales"].sum())

# j) Sort by Total_Sales in descending order
df = df.sort_values(by="Total_Sales", ascending=False)
print("\nj) Sorted Sales Data:")
print(df)

# k) Save updated DataFrame
df.to_csv("sales_analysis.csv", index=False)
print("\nk) Data saved to sales_analysis.csv")
```

IDLE Shell 3.13.15

File Edit Shell Debug Options Window Help

```
Python 3.13.15 (tags/v3.13.15:406lbc4, Aug  5 2026, 13:05:39) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: E:/107 Lab task/python/Pandas/sales_analysis.py ==========
b) All Products:
     Product     Category  Price  Quantity
0     Laptop  Electronics  50000         5
1     Mobile  Electronics  20000        15
2      Mouse  Accessories    500        25
3   Keyboard  Accessories   1000        12
4    Monitor  Electronics  12000         8
5     Tablet  Electronics  15000        10
6    Printer  Electronics   8000         6
7  Headphone  Accessories   1500        20
8    Speaker  Accessories   2500        14
9    Charger  Accessories    600        30

c) and d) Total Sales:
     Product  Total_Sales
0     Laptop       250000
1     Mobile       300000
2      Mouse        12500
3   Keyboard        12000
4    Monitor        96000
5     Tablet       150000
6    Printer        48000
7  Headphone        30000
8    Speaker        35000
9    Charger        18000

e) Product with Highest Sales:
Product             Mobile
Category       Electronics
Price                20000
Quantity                15
Total_Sales         300000
Name: 1, dtype: object

f) Total Sales:
951500

g) Average Product Price:
11110.0

h) Quantity Greater Than 10:
     Product     Category  Price  Quantity  Total_Sales
1     Mobile  Electronics  20000        15       300000
2      Mouse  Accessories    500        25        12500
3   Keyboard  Accessories   1000        12        12000
7  Headphone  Accessories   1500        20        30000
8    Speaker  Accessories   2500        14        35000
9    Charger  Accessories    600        30        18000

i) Category-wise Total Sales:
Category
Accessories    107500
Electronics    844000
Name: Total_Sales, dtype: int64
```

```
g) Average Product Price:
11110.0

h) Quantity Greater Than 10:
     Product     Category  Price  Quantity  Total_Sales
1     Mobile  Electronics  20000        15       300000
2      Mouse  Accessories    500        25        12500
3   Keyboard  Accessories   1000        12        12000
7  Headphone  Accessories   1500        20        30000
8    Speaker  Accessories   2500        14        35000
9    Charger  Accessories    600        30        18000

i) Category-wise Total Sales:
Category
Accessories    107500
Electronics    844000
Name: Total_Sales, dtype: int64

j) Sorted Sales Data:
     Product     Category  Price  Quantity  Total_Sales
1     Mobile  Electronics  20000        15       300000
0     Laptop  Electronics  50000         5       250000
5     Tablet  Electronics  15000        10       150000
4    Monitor  Electronics  12000         8        96000
6    Printer  Electronics   8000         6        48000
8    Speaker  Accessories   2500        14        35000
7  Headphone  Accessories   1500        20        30000
9    Charger  Accessories    600        30        18000
2      Mouse  Accessories    500        25        12500
3   Keyboard  Accessories   1000        12        12000

k) Data saved to sales_analysis.csv
>>>
```

12.Missing Data Handling

Create a CSV file named employee_data.csv
containing: EmpID, Name, Age, Department, Salary
Include some missing values in the CSV file.

Using Pandas, perform the following:

a) Read the CSV file.
b) Display the complete dataset.
c) Check whether the dataset contains missing values.
d) Count the number of missing values in each column.
e) Replace missing Age values with the average age.
f) Replace missing Salary values with 0.
g) Replace missing Department values with "Not Assigned".
h) Display the updated DataFrame.
i) Sort employees according to salary.
j) Save the cleaned data into cleaned_employee_data.csv.

missing_data.py - E:/107 Lab task/python/Pandas/missing_data.py (3.13.15)

File Edit Format Run Options Window Help

```
import pandas as pd

data = {
    "EmpID": [101, 102, 103, 104, 105, 106, 107, 108, 109, 110],
    "Name": ["Amit", "Neha", "Rahul", "Priya", "Sneha",
             "Rohit", "Pooja", "Akash", "Kiran", "Sahil"],
    "Age": [25, 30, None, 28, 35, None, 26, 32, 29, 31],
    "Department": ["IT", "HR", None, "Sales", "IT",
                   "Finance", None, "HR", "Sales", "IT"],
    "Salary": [30000, 40000, 35000, None, 50000,
               45000, None, 38000, 42000, 55000]
}

df = pd.DataFrame(data)
df.to_csv("employee_data.csv", index=False)

df = pd.read_csv("employee_data.csv")

print("b) Complete Dataset:")
print(df)

print("\nc) Missing Values Present:")
print(df.isnull().values.any())

print("\nd) Missing Values Count:")
print(df.isnull().sum())

df["Age"] = df["Age"].fillna(df["Age"].mean())

df["Salary"] = df["Salary"].fillna(0)

df["Department"] = df["Department"].fillna("Not Assigned")

print("\nh) Updated DataFrame:")
print(df)
df = df.sort_values(by="Salary")

print("\ni) Employees Sorted by Salary:")
print(df)
df.to_csv("cleaned_employee_data.csv", index=False)

print("\nj) Cleaned data saved successfully!")
```

Output:

IDLE Shell 3.13.15

File Edit Shell Debug Options Window Help

```
Python 3.13.15 (tags/v3.13.15:4061bc4, Aug  5 2026, 13:05:39) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
============ RESTART: E:/107 Lab task/python/Pandas/missing_data.py ============
b) Complete Dataset:
   EmpID   Name   Age Department   Salary
0    101   Amit  25.0         IT  30000.0
1    102   Neha  30.0         HR  40000.0
2    103  Rahul   NaN        NaN  35000.0
3    104  Priya  28.0      Sales      NaN
4    105  Sneha  35.0         IT  50000.0
5    106  Rohit   NaN    Finance  45000.0
6    107  Pooja  26.0        NaN      NaN
7    108  Akash  32.0         HR  38000.0
8    109  Kiran  29.0      Sales  42000.0
9    110  Sahil  31.0         IT  55000.0

c) Missing Values Present:
True

d) Missing Values Count:
EmpID         0
Name          0
Age           2
Department    2
Salary        2
dtype: int64

h) Updated DataFrame:
   EmpID   Name   Age    Department   Salary
0    101   Amit  25.0            IT  30000.0
1    102   Neha  30.0            HR  40000.0
2    103  Rahul  29.5  Not Assigned  35000.0
3    104  Priya  28.0         Sales      0.0
4    105  Sneha  35.0            IT  50000.0
5    106  Rohit  29.5       Finance  45000.0
6    107  Pooja  26.0  Not Assigned      0.0
7    108  Akash  32.0            HR  38000.0
8    109  Kiran  29.0         Sales  42000.0
9    110  Sahil  31.0            IT  55000.0

i) Employees Sorted by Salary:
   EmpID   Name   Age    Department   Salary
3    104  Priya  28.0         Sales      0.0
6    107  Pooja  26.0  Not Assigned      0.0
0    101   Amit  25.0            IT  30000.0
2    103  Rahul  29.5  Not Assigned  35000.0
7    108  Akash  32.0            HR  38000.0
1    102   Neha  30.0            HR  40000.0
8    109  Kiran  29.0         Sales  42000.0
5    106  Rohit  29.5       Finance  45000.0
4    105  Sneha  35.0            IT  50000.0
9    110  Sahil  31.0            IT  55000.0

j) Cleaned data saved successfully!
>>>
```

13.College Placement Analysis

Create a CSV file named placement.csv containing:

Name, Branch, Company, Package, Status

Store data for at least 10 students, including both selected and not-selected students.

Using Pandas, perform the following:

a) Read the CSV file.
b) Display all student records.
c) Display only selected students.
d) Count the total number of selected students.
e) Find the highest package.
f) Find the lowest package among selected students.
g) Calculate the average package of selected students.
h) Display students having a package greater than 4 LPA.
i) Sort selected students according to package in descending order.
j) Find the number of students selected by each company using groupby().
k) Save the selected-student data into selected_students.csv.

placement_analysis.py - E:/107 Lab task/python/Pandas/placement_analysis.py (3.13.15)

File Edit Format Run Options Window Help

```
import pandas as pd

data = {
    "Name": ["Amit", "Neha", "Rahul", "Priya", "Sneha",
             "Rohit", "Pooja", "Akash", "Kiran", "Sahil"],
    "Branch": ["IT", "CS", "ENTC", "IT", "CS",
               "ENTC", "IT", "CS", "ENTC", "IT"],
    "Company": ["TCS", "Infosys", "Wipro", "Google", "TCS",
                "Infosys", "Wipro", "Google", "TCS", "Infosys"],
    "Package": [4.5, 3.5, 0, 8.0, 5.0, 0, 4.0, 6.0, 0, 3.0],
    "Status": ["Selected", "Selected", "Not Selected",
               "Selected", "Selected", "Not Selected",
               "Selected", "Selected", "Not Selected", "Selected"]
}

df = pd.DataFrame(data)
df.to_csv("placement.csv", index=False)


df = pd.read_csv("placement.csv")

print("b) All Student Records:")
print(df)


selected = df[df["Status"] == "Selected"]
print("\nc) Selected Students:")
print(selected)


print("\nd) Total Selected Students:")
print(len(selected))


print("\ne) Highest Package:")
print(selected["Package"].max(), "LPA")

print("\nf) Lowest Package:")
print(selected["Package"].min(), "LPA")

print("\ng) Average Package:")
print(round(selected["Package"].mean(), 2), "LPA")

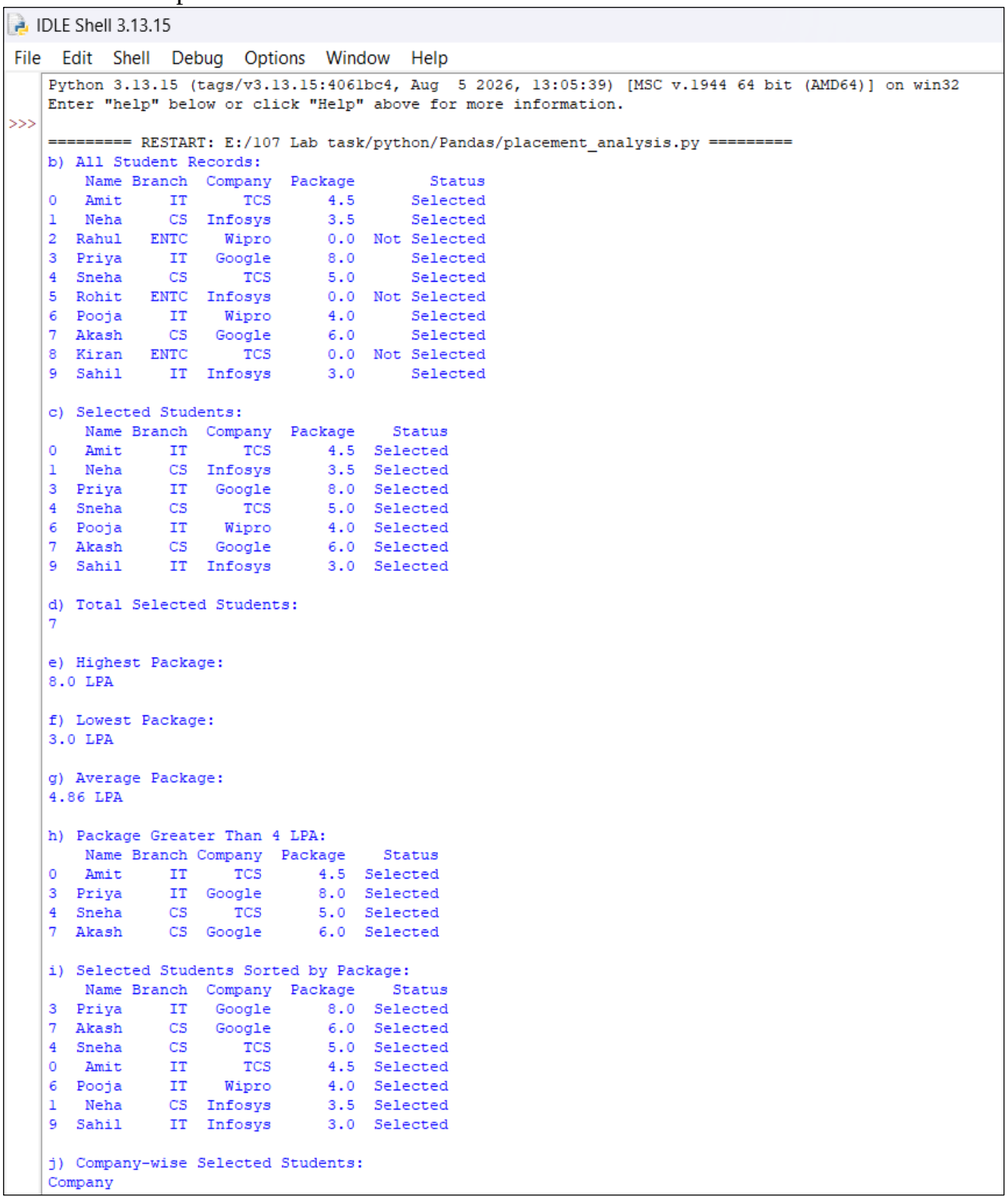
print("\nh) Package Greater Than 4 LPA:")
print(selected[selected["Package"] > 4])

selected = selected.sort_values(by="Package", ascending=False)
print("\ni) Selected Students Sorted by Package:")
print(selected)

print("\nj) Company-wise Selected Students:")
print(selected.groupby("Company")["Name"].count())

selected.to_csv("selected_students.csv", index=False)
print("\nk) Data saved to selected_students.csv")
```

Output:


IDLE Shell 3.13.15
File Edit Shell Debug Options Window Help
Python 3.13.15 (tags/v3.13.15:4061bc4, Aug 5 2026, 13:05:39) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========= RESTART: E:/107 Lab task/python/Pandas/placement_analysis.py =========
b) All Student Records:
Name Branch Company Package Status
0 Amit IT TCS 4.5 Selected
1 Neha CS Infosys 3.5 Selected
2 Rahul ENTC Wipro 0.0 Not Selected
3 Priya IT Google 8.0 Selected
4 Sneha CS TCS 5.0 Selected
5 Rohit ENTC Infosys 0.0 Not Selected
6 Pooja IT Wipro 4.0 Selected
7 Akash CS Google 6.0 Selected
8 Kiran ENTC TCS 0.0 Not Selected
9 Sahil IT Infosys 3.0 Selected
c) Selected Students:
Name Branch Company Package Status
0 Amit IT TCS 4.5 Selected
1 Neha CS Infosys 3.5 Selected
3 Priya IT Google 8.0 Selected
4 Sneha CS TCS 5.0 Selected
6 Pooja IT Wipro 4.0 Selected
7 Akash CS Google 6.0 Selected
9 Sahil IT Infosys 3.0 Selected
d) Total Selected Students:
7
e) Highest Package:
8.0 LPA
f) Lowest Package:
3.0 LPA
g) Average Package:
4.86 LPA
h) Package Greater Than 4 LPA:
Name Branch Company Package Status
0 Amit IT TCS 4.5 Selected
3 Priya IT Google 8.0 Selected
4 Sneha CS TCS 5.0 Selected
7 Akash CS Google 6.0 Selected
i) Selected Students Sorted by Package:
Name Branch Company Package Status
3 Priya IT Google 8.0 Selected
7 Akash CS Google 6.0 Selected
4 Sneha CS TCS 5.0 Selected
0 Amit IT TCS 4.5 Selected
6 Pooja IT Wipro 4.0 Selected
1 Neha CS Infosys 3.5 Selected
9 Sahil IT Infosys 3.0 Selected
j) Company-wise Selected Students:
Company

```
h) Package Greater Than 4 LPA:
    Name Branch Company  Package    Status
0   Amit     IT     TCS      4.5  Selected
3  Priya     IT  Google      8.0  Selected
4  Sneha     CS     TCS      5.0  Selected
7  Akash     CS  Google      6.0  Selected

i) Selected Students Sorted by Package:
    Name Branch  Company  Package    Status
3  Priya     IT   Google      8.0  Selected
7  Akash     CS   Google      6.0  Selected
4  Sneha     CS      TCS      5.0  Selected
0   Amit     IT      TCS      4.5  Selected
6  Pooja     IT    Wipro      4.0  Selected
1   Neha     CS  Infosys      3.5  Selected
9  Sahil     IT  Infosys      3.0  Selected

j) Company-wise Selected Students:
Company
Google     2
Infosys    2
TCS        2
Wipro      1
Name: Name, dtype: int64

k) Data saved to selected_students.csv
>
```

14.Department Analysis

Create a DataFrame containing:

Name, Department, Salary, Experience, City

Using Pandas, perform the following:

a) Display employees with salary > ₹50,000.
b) Find the average salary.
c) Find the highest salary.
d) Find employees with more than 3 years of experience.
e) Count employees city-wise.
f) Calculate average salary department-wise.
g) Sort employees by salary.
h) Save the final DataFrame as a CSV file.

department_analysis.py - E:/107 Lab task/python/Pandas/department_analysis.py

File Edit Format Run Options Window Help

```python
import pandas as pd


data = {
    "Name": ["Amit", "Neha", "Rahul", "Priya", "Sneha",
             "Rohit", "Pooja", "Akash", "Kiran", "Sahil"],
    "Department": ["IT", "HR", "IT", "Sales", "HR",
                   "Finance", "IT", "Sales", "Finance", "HR"],
    "Salary": [55000, 40000, 60000, 45000, 35000,
               52000, 48000, 65000, 42000, 38000],
    "Experience": [4, 2, 6, 3, 1, 5, 4, 7, 2, 3],
    "City": ["Pune", "Mumbai", "Pune", "Kolhapur", "Mumbai",
             "Pune", "Kolhapur", "Mumbai", "Pune", "Kolhapur"]
}

df = pd.DataFrame(data)


print("a) Salary Greater Than 50000:")
print(df[df["Salary"] > 50000])


print("\nb) Average Salary:")
print(df["Salary"].mean())

print("\nc) Highest Salary:")
print(df["Salary"].max())

print("\nd) Experience More Than 3 Years:")
print(df[df["Experience"] > 3])

print("\ne) City-wise Employee Count:")
print(df.groupby("City")["Name"].count())

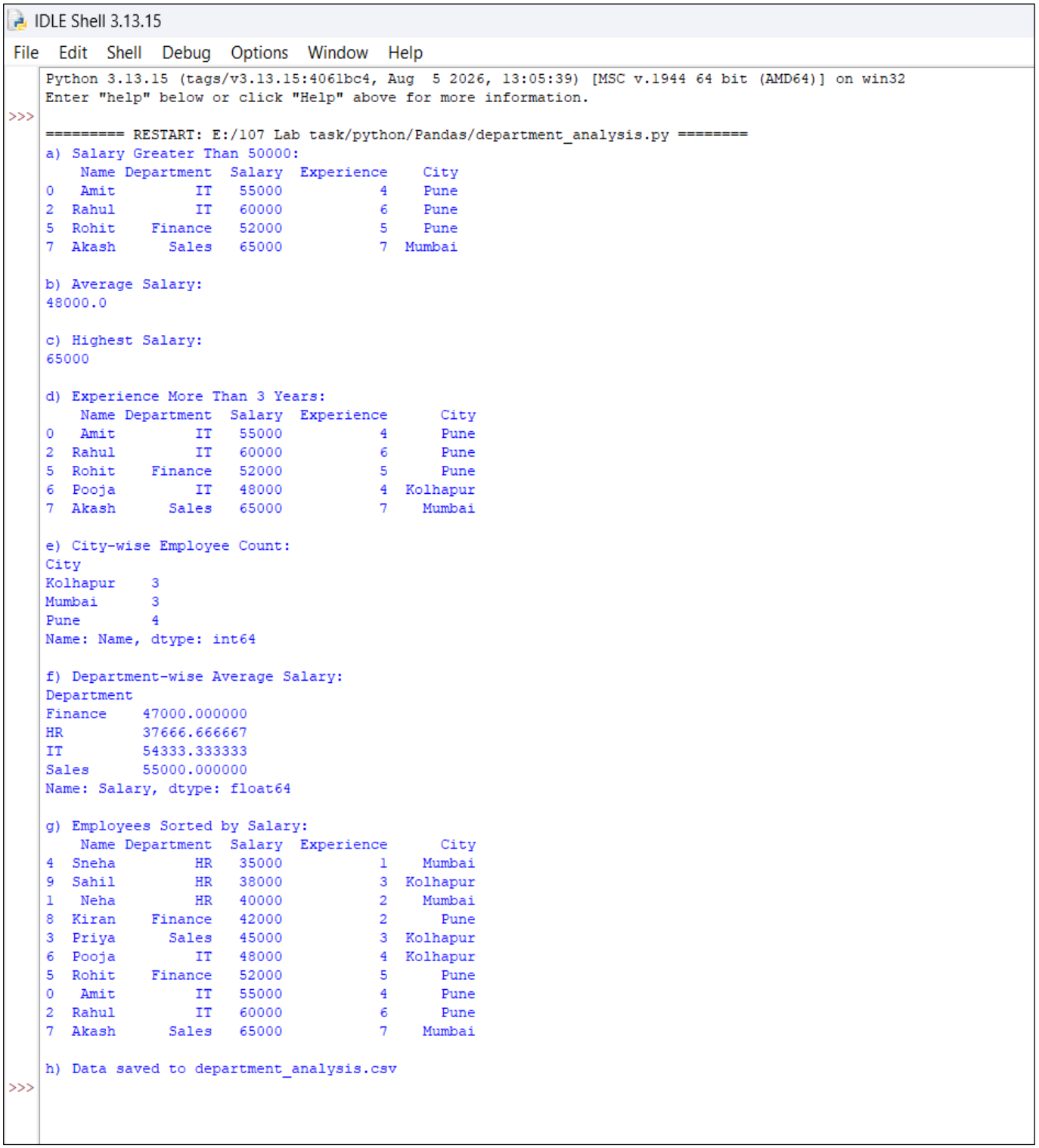
print("\nf) Department-wise Average Salary:")
print(df.groupby("Department")["Salary"].mean())


df = df.sort_values(by="Salary")

print("\ng) Employees Sorted by Salary:")
print(df)

df.to_csv("department_analysis.csv", index=False)
print("\nh) Data saved to department_analysis.csv")
```

Output:


IDLE Shell 3.13.15
File Edit Shell Debug Options Window Help
Python 3.13.15 (tags/v3.13.15:4061bc4, Aug 5 2026, 13:05:39) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========= RESTART: E:/107 Lab task/python/Pandas/department_analysis.py ========
a) Salary Greater Than 50000:
Name Department Salary Experience City
0 Amit IT 55000 4 Pune
2 Rahul IT 60000 6 Pune
5 Rohit Finance 52000 5 Pune
7 Akash Sales 65000 7 Mumbai
b) Average Salary:
48000.0
c) Highest Salary:
65000
d) Experience More Than 3 Years:
Name Department Salary Experience City
0 Amit IT 55000 4 Pune
2 Rahul IT 60000 6 Pune
5 Rohit Finance 52000 5 Pune
6 Pooja IT 48000 4 Kolhapur
7 Akash Sales 65000 7 Mumbai
e) City-wise Employee Count:
City
Kolhapur 3
Mumbai 3
Pune 4
Name: Name, dtype: int64
f) Department-wise Average Salary:
Department
Finance 47000.000000
HR 37666.666667
IT 54333.333333
Sales 55000.000000
Name: Salary, dtype: float64
g) Employees Sorted by Salary:
Name Department Salary Experience City
4 Sneha HR 35000 1 Mumbai
9 Sahil HR 38000 3 Kolhapur
1 Neha HR 40000 2 Mumbai
8 Kiran Finance 42000 2 Pune
3 Priya Sales 45000 3 Kolhapur
6 Pooja IT 48000 4 Kolhapur
5 Rohit Finance 52000 5 Pune
0 Amit IT 55000 4 Pune
2 Rahul IT 60000 6 Pune
7 Akash Sales 65000 7 Mumbai
h) Data saved to department_analysis.csv
>>>